Algebraic expressions

1. Simplify

a) $4m^2n + 5mn^2 - 2m^2n + mn^2 - 3mn^2$

b) $3x^2 - 5x + 2 + 3x^2 - 7x - 12$

1. Simplify

2. Single power of 2

a) $2^5 \times 2^3$ b) $2^6 \div 2^2$ c) $(2^3)^2$

2. Single power of 2

3. Expand

a) $3(x + 4)$ b) $5(2 - 3x)$ c) $6(2x - 5y)$

3. Expand

4. Highest common factor

a) $24, 16$ b) $6x, 8x^2$ c) $4xy^2, 3xy$

4. Highest common factor

5. Simplify

a) $\frac{10x}{5}$ b) $\frac{20x}{2}$ c) $\frac{40x}{24}$

5. Simplify

1: Basic Index Laws

2: Expand brackets

3: Factorise quadratics/cubics

4: Fractional/Negative Powers

5: Surds

1: Index Laws

In 3^5, 3 is the base and 5 is the exponent (or index). The whole expression is a power.

Use precise language: when multiplying powers with the same base, add the indices.

$$a^m a^n = a^{m+n}$$
$$a^m / a^n = a^{m-n}$$
$$(a^m)^n = a^{mn}$$
$$(ab)^n = a^n b^n$$

1 Simplify $(a^3)^2 \times 2a^2$.

1

2 Simplify $(4x^3y)^3$.

2

3 Simplify $2x^2(3+5x) - x(4-x^2)$.

3

4 Simplify $\frac{x^3-2x}{3x^2}$.

4

Test Your Understanding 5

1 Simplify

$$\left(\frac{2a^5}{a^2}\right)^2 \times 3a$$

1

2 Simplify

$$\frac{2x + x^5}{4x^3}$$

2

3 Expand and simplify

$$2x(3 - x^2) - 4x^3(3 - x)$$

3

4 Simplify

$$2^x \times 3^x$$

4

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Exercise 2: expand and simplify. Exercise 3: simplify each fraction.

2(o)

$x(3x^2 - 2x + 5)$

2(o)

2(p)

$7y^2(2 - 5y + 3y^2)$

2(p)

2(q)

$-2y^2(5 - 7y + 3y^2)$

2(q)

2(r)

$7(x - 2) + 3(x + 4) - 6(x - 2)$

2(r)

2(s)

$5x - 3(4 - 2x) + 6$

2(s)

2(t)

$3x^2 - x(3 - 4x) + 7$

2(t)

2(u)

$4x(x + 3) - 2x(3x - 7)$

2(u)

2(v)

$3x^2(2x + 1) - 5x^2(3x - 4)$

2(v)

3(d)

$$\frac{8x^3 + 5x}{2x}$$

3(d)

3(e)

$$\frac{7x^7 + 5x^2}{5x}$$

3(e)

3(f)

$$\frac{9x^5 - 5x^3}{3x}$$

3(f)

1. MAT 2006 1A Which number is largest?

A) $((2^3)^2)^3$ B) $(2^3)^{2^3}$

C) $2^{(3^2)^3}$ D) $2^{3^{2^3}}$

1. MAT 2006 1A

2. MAT 2012 1B Let $N = 2^k 4^m 8^n$, where k, m, n are positive integers. When is N definitely a square?

A) k even B) $k + n$ odd

C) k odd and $m + n$ even D) $k + n$ even

2. MAT 2012 1B

Multiply each term in the first bracket by every term in the second bracket. Work systematically.

1

$(x - y)(x + y - 1)$

1

2

$(x + 1)(x + 2)(x + 3)$

2

For three or more brackets, expand two at a time.

1 Expand $(x+5)(x-2)(x+1)$.

1

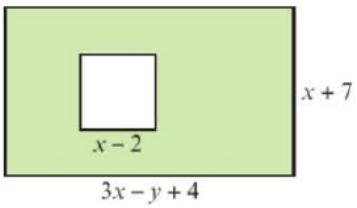
2 Expand $2(x-3)(x-4)$.

2

3 Expand $(2x-1)^3$.

3

3 The diagram shows a rectangle with a square cut out.
The rectangle has length $3x - y + 4$ and width $x + 7$.
The square has length $x - 2$.
Find an expanded and simplified expression for the shaded area.



3

5 Find a, b, c, d when

$(2x+5y)(3x-y)(2x+y) = ax^3 + bx^2y + cxy^2 + dy^3$

5

1. MAT 2002 1B Which identities are correct?

(i) $yz(z-y)+zx(x-z)+xy(y-x)$
$=(z-y)(x-z)(y-x)$

(ii) $yz(z-y)+zx(x-z)+xy(y-x)$
$=(z-y)(z-x)(y-x)$

(iii) $yz(x+y)+zx(z+x)+xy(y+x)$
$=(z+y)(z+x)(y+x)$

1. MAT 2002 1B

2. MAT 2007 1E For integers x, n, consider
$(1-x)^n(2-x)^{2n}(3-x)^{3n}(4-x)^{4n}(5-x)^{5n}$
Which statement holds?

A. Negative when $n>5$ and $x<5$.

B. Negative when n is odd and $x>5$.

C. Negative when n is a multiple of 3 and $x>5$.

D. Negative when n is even and $x<5$.

2. MAT 2007 1E

3: Factorising 12

Factorising reverses expanding: write an expression as a product of factors.

$x(x+1)(x+2)$

Factorised: a product of three linear factors. A linear expression has the form $ax+b$.

$x(x+1)+(x-1)(x+1)$

Not yet factorised: the outermost operation is addition.

Basic examples

1 $x^3+x^2=$

1

2 $4x-8xy=$

2

Factorising Quadratics 13

Recap

For $x^2 + bx + c$, find two numbers that add to b and multiply to c. The coefficient is the number multiplying a term.

1 $x^2 - 5x - 14 =$

1

When the coefficient of x^2 is not 1

Guess and check the brackets, or split the middle term.

2 $2x^2 + 5x - 12 =$

2

Split the middle term Find two numbers with sum 5 and product $2(-12) = -24$.

Split the middle term

Working Use those numbers to split $5x$, then factorise both pairs.

Working

Other Factorisations 14

Difference of two squares

$a^2 - b^2 = (a + b)(a - b)$. Always look for a common factor first.

1 $4x^2 - 9 =$

1

2 $x^3 - x =$

2

3 $x^3 + 3x^2 + 2x =$

3

Test Your Understanding 15

1. Factorise completely

$6x^2 + x - 2$

1. Factorise completely

2. Factorise completely

$x^3 - 7x^2 + 12x$

2. Factorise completely

3. Challenge Factorise

$x^4 - 1$

3. Challenge

4. Challenge Factorise

$x^3 - 1$

4. Challenge

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4 Factorise $x^4 - y^4$.

4

5 Factorise $6x^3 + 7x^2 - 5x$.

5

Challenge Write $4x^4 - 13x^2 + 9$ as four linear factors.

Challenge

4: Negative and Fractional Indices

$a^0 = 1, \quad a^{1/m} = \sqrt[m]{a}, \quad a^{n/m} = (\sqrt[m]{a})^n, \quad a^{-m} = \dfrac{1}{a^m}$

Use positive a for these fractional-index laws.

$\sqrt{9} = 3$, the positive square root; $\pm\sqrt{9}$ gives both 3 and -3.

1 Explain why $x^{1/2} = \sqrt{x}$ for $x > 0$.

1

2 Evaluate $27^{-1/3}$.

2

3 Evaluate $32^{2/5}$.

3

4 Simplify $(\frac{1}{9}x^6y)^{1/2}$, assuming $x, y > 0$.

4

5 Evaluate $(\frac{27}{8})^{-2/3}$.

5

6 If $b = \frac{1}{9}a^2$, express $3b^{-2}$ in the form ka^n.

6

Writing a surd using indices

Example If $9\sqrt{3} = 3^k$, find k.

Example

1. Edexcel IAL C12 Jan 2019 Q2(c) Given $y = 2^x$, express $1/4^{2x-3}$ in terms of y.

1. Edexcel IAL C12 Jan 2019 Q2(c)

2. Edexcel IGCSE May 2014 4H Given $(2^{1/2})^n = 2^x/8^y$, express n in terms of x, y.

2. Edexcel IGCSE May 2014 4H

Question Given $2^x 4^y = 1/(2\sqrt{2})$, express y as a function of x. (3 marks)

Question

4. Evaluate or simplify

4(a) $81^{1/4} =$

4(a)

4(b) $x(2x^{-1/3})^4 =$

4(b)

5. Given $y = \frac{1}{8}x^3$, express in terms of x.

5(a) $y^{1/3} =$

5(a)

5(b) $\frac{1}{2}y^{-2} =$

5(b)

Challenge · MAT 2007 1A For integers r, s, when is this expression an integer?

$$\frac{6^{r+s}12^{r-s}}{8^r 9^{r+2s}}$$

Choose: $r + s \leq 0$, $s \leq 0$, $r \leq 0$, or $r \geq s$.

Challenge · MAT 2007 1A

A surd is a root that does not simplify to a rational number. A rational number can be written a/b, where a, b are integers and $b \neq 0$.

$\sqrt{a}\sqrt{b} = \sqrt{ab}$, $\quad \frac{\sqrt{a}}{\sqrt{b}} = \sqrt{\frac{a}{b}} \quad (a \geq 0, b > 0)$

1 $\sqrt{3} \times 2 =$

1

2 $3\sqrt{5} \times 2\sqrt{5} =$

2

3 $\sqrt{8} =$

3

4 $\sqrt{12} + \sqrt{27} =$

4

5 $(\sqrt{8} + 1)(\sqrt{2} - 3) =$

5

3 Simplify $\sqrt{75} - \sqrt{12}$ in the form $a\sqrt{3}$, with a an integer.

3

Extension

1. SMC 2014 Q24 Which is smallest?

$10 - 3\sqrt{11}$; $8 - 3\sqrt{7}$; $5 - 2\sqrt{6}$;
$9 - 4\sqrt{5}$; $7 - 4\sqrt{3}$

1. SMC 2014 Q24

2. SMC 2012 Q21 Which has no square root of the form $x + y\sqrt{2}$, with positive integers x, y?

$17 + 12\sqrt{2}$; $22 + 12\sqrt{2}$; $38 + 12\sqrt{2}$;
$54 + 12\sqrt{2}$; $73 + 12\sqrt{2}$

2. SMC 2012 Q21

6: Rationalising The Denominator 24

What can we multiply this surd by to obtain a rational number?

$\sqrt{5}\times$ Multiplier 1 $=$ Result 1

Rationalising a denominator makes it rational. Multiply the numerator and denominator by the same non-zero expression, so the value stays unchanged.

$\frac{1}{\sqrt{2}}\times$ Multiplier 2 $=$ Result 2

This gives a simpler denominator and makes expressions involving surds easier to combine.

Worked examples · Rationalise the denominator

1 $\frac{3}{\sqrt{2}} =$ [1]

2 $\frac{6}{\sqrt{3}} =$ [2]

3 $\frac{7}{\sqrt{7}} =$ [3]

4 $\frac{15}{\sqrt{5}} + \sqrt{5} =$ [4]

Test your understanding

5
$\frac{12}{\sqrt{3}}$
[5]

6
$\frac{2}{\sqrt{6}}$
[6]

7
$\frac{4\sqrt{2}}{\sqrt{8}}$
[7]

More Complex Denominators 26

For a denominator with two terms, multiply by its conjugate: reverse the sign between the terms.

$\frac{1}{\sqrt{2}+1} \times$ [1. Conjugate]

$=$ [2. Simplify]

Why this works

$(a+b)(a-b) = a^2 - b^2$

The difference of two squares removes the surd from the denominator.

1 Rationalise

$$\frac{3}{\sqrt{6}-2}$$

1

2 Rationalise

$$\frac{4}{\sqrt{3}+1}$$

2

3 Rationalise

$$\frac{3\sqrt{2}+4}{5\sqrt{2}-7}$$

3

Remember $(a-b)(a+b)=a^2-b^2$: use the difference of two squares to simplify each denominator.

1. AQA IGCSE FM June 2013 Paper 1 Solve $y(\sqrt{3}-1)=8$. Give y in the form $a+b\sqrt{3}$.

1. AQA IGCSE FM June 2013 Paper 1

2 Rationalise and simplify $\frac{2\sqrt{3}-1}{3\sqrt{3}+1}$.

2

3 Rationalise and simplify $\frac{4}{\sqrt{5}-2}$.

3

Writing surds in power form is useful in later calculus questions.

Example Express $\frac{(2+\sqrt{x})^2}{x}$ as powers of x.

Example

Test your understanding Express $\frac{(x+\sqrt{x})^2}{2\sqrt{x}}$ as powers of x.

Test your understanding

1. Rationalise and simplify

1(a) $\frac{1}{\sqrt{5}+2} =$

1(a)

1(b) $\frac{\sqrt{3}}{\sqrt{3}-1} =$

1(b)

1(c) $\frac{\sqrt{5}+1}{\sqrt{5}-2} =$

1(c)

1(d) $\frac{2\sqrt{3}-1}{3\sqrt{3}+4} =$

1(d)

1(e) $\frac{5\sqrt{5}-2}{2\sqrt{5}-3} =$

1(e)

2 Expand $(\sqrt{5}+3)(\sqrt{5}-2)(\sqrt{5}+1)$.

2

3 Rationalise $\frac{3\sqrt{3}+7}{3\sqrt{3}-5}$ in the form $a+b\sqrt{3}$.

3

4 Solve $x(4-\sqrt{6}) = 10$.

4

5 Solve $y(1+\sqrt{2}) - \sqrt{2} = 3$.

5

6 Simplify $\frac{\sqrt{a+1}-\sqrt{a}}{\sqrt{a+1}+\sqrt{a}}$.

6

Challenge Solve $\frac{\sqrt[4]{9}}{\sqrt[5]{27}} = \sqrt[x]{3}$, interpreting the right side as $3^{1/x}$.

Challenge